By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

Macaulay Beam Analysis - BetaTesting

06 - Steel - Triangular Distributed Load (1)

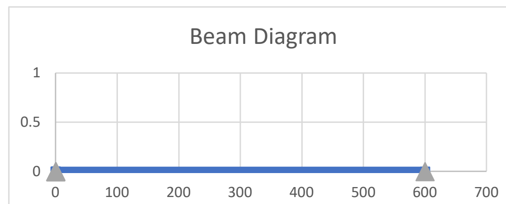
Bridge, L =		in	E =	29,000,000	lb / in^2	Length Units	in
Beam Sect =	W14x61		G =	1.740E+06	lb / in^2 = 0.06*29,000,000	Force Units	lb
A =	17.90	in^2					
I =	640.00	in^4					

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in^2	in^4	lb / in^2	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	dY (vert)	dr (rot)
			Perscribed Disp. (typ 0)	
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			

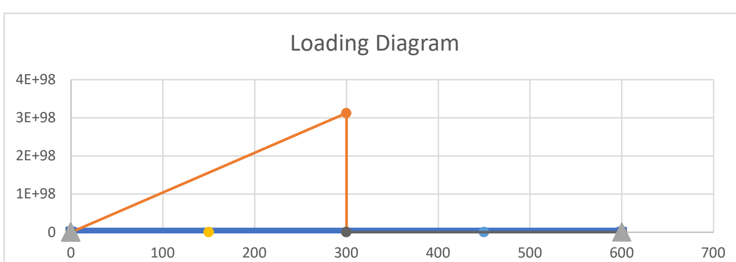


Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	300.0	0.0	-6.3	75.0
	2	300.0	600.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	150	0.00	0.00
	2	450	0.00	0.00
	3			
	4			
	5			



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Macaulay Test

BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	625.0	0.0	Support reaction
600.00	0.00	0.00	312.5	0.0	Support reaction

Vert Reac
(kip)
0.625
0.313

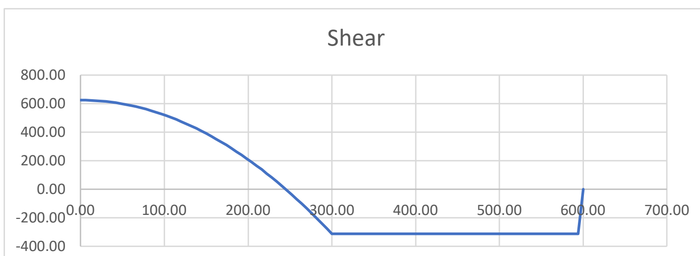
BEAM REACTIONS EQUILIBRUM CHECK

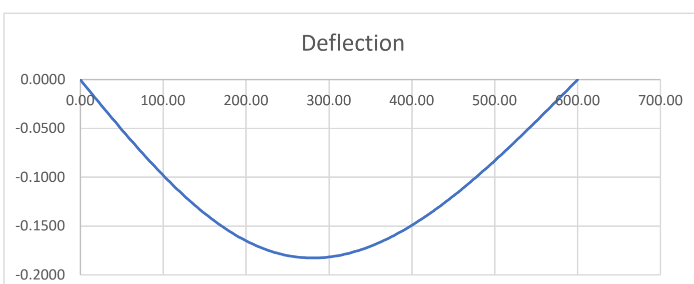
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-937.5	937.5	0.0	OK
Sum M @ x=0	F-L	-1.8750E+05	1.8750E+05	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Value
(---)	(L)	(F or F-L)
Max +V	0.00	625.0
Max -V	336.00	-312.5
Max +M	246.00	102,059.3
Max -M	0.00	0.0
Max Defl	282.00	-0.183

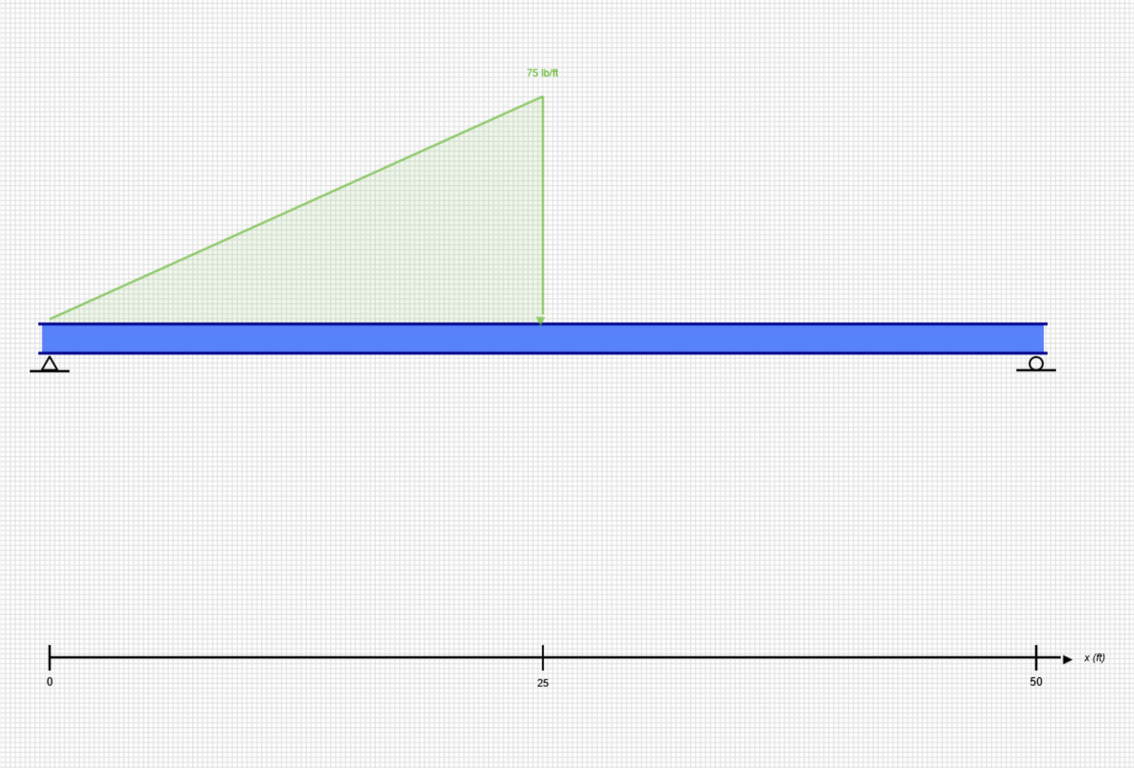
Value	
(---)	
0.63	kip
-0.31	kip
8.50	kip-ft
0.00	kip-ft
-0.183	in





SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Tue Jun 09 2026 14:08:40 GMT-0400 (Eastern Daylight Time)

Project Info

File Name: 01-Steel-Beam

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

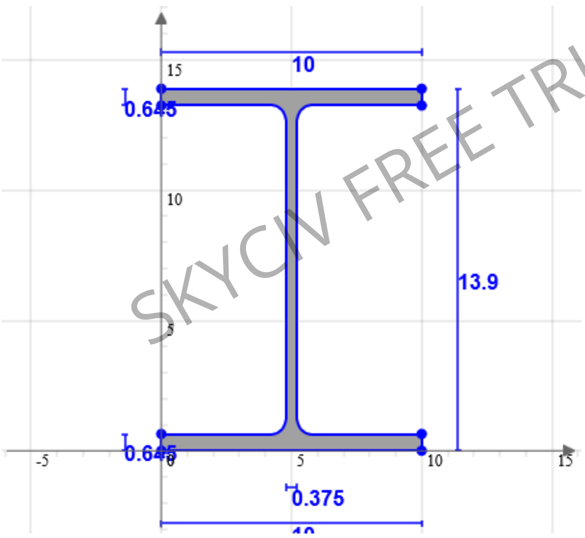
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft

Loads

Load Type	Location	Magnitude	Load Case
Distributed Load	0 ft to 25 ft	0 lb to -75 lb	DL

Beam Section



Geometric Properties		
A	17.9	in^2
C_z	5	in
C_y	6.95	in

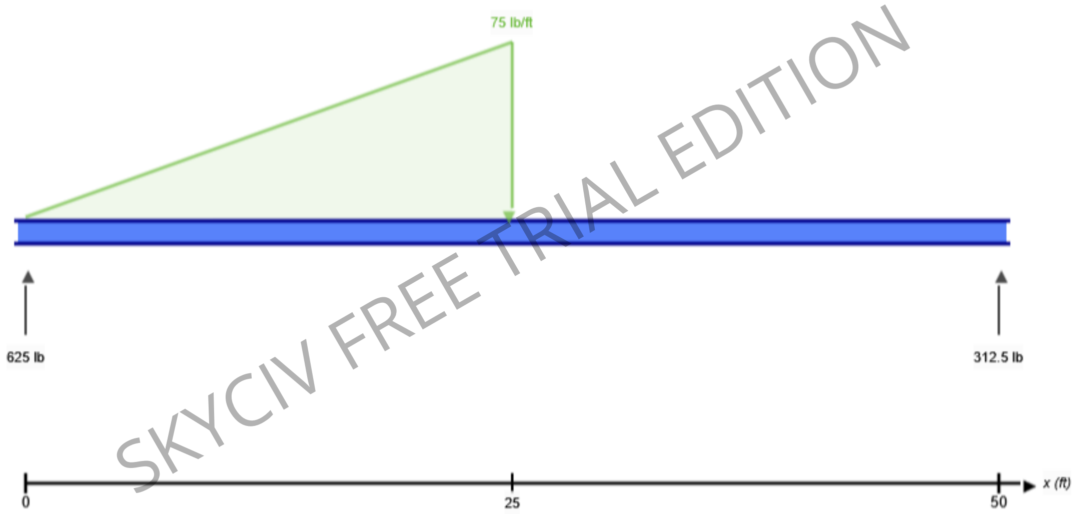
Bending Properties		
I_z	640	in^4
I_y	107	in^4

Shear Properties		
A_z	11.277	in^2
A_y	4.99	in^2

Torsion Properties		
J	2.19	in^4
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft^3)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.076	L/3284
Custom Stress Limit	PASS	39 ksi	0.028	1.108 ksi
Material Yield	PASS	36 ksi	0.031	1.108 ksi
Material Strength	PASS	58 ksi	0.019	1.108 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	625	0
50	0	312.5	0

Force Extremes

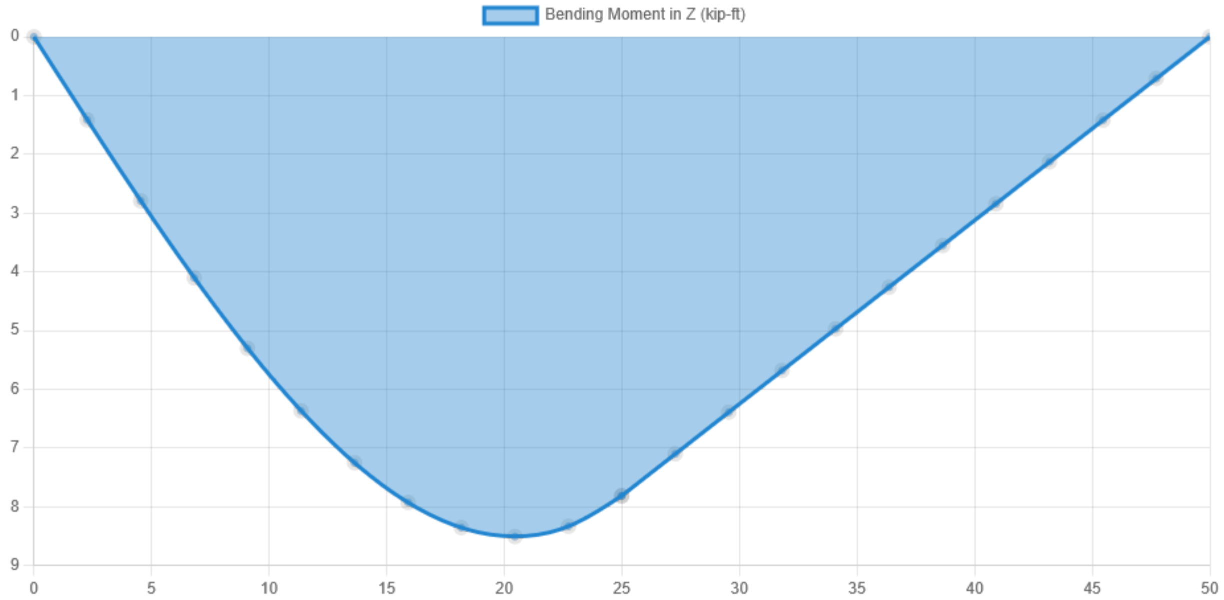
Result	Max	Min
Bending Moment	8.505 kip-ft	0 kip-ft
Shear	625 lb	-312.5 lb
Displacement Y	0 in	-0.183 in

Stress Extremes

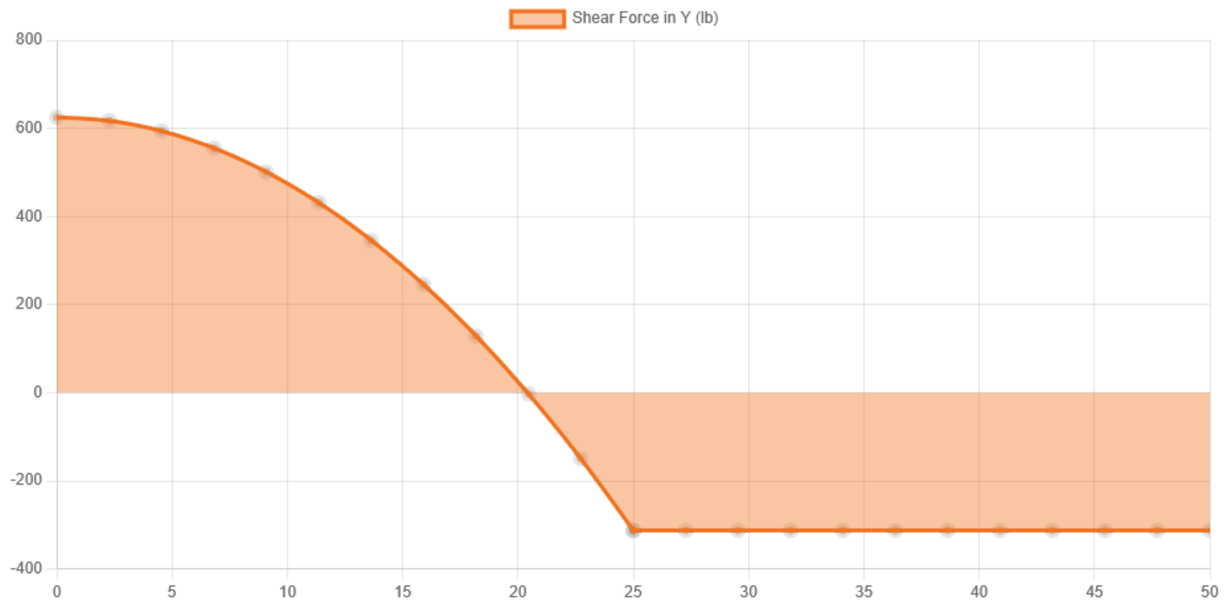
Result	Max (ksi)	Min (ksi)
Bending Stress	1.108	-1.108
Shear Stress Total	0.133	0.001
Max Combined Normal Stress	1.108	0
Min Combined Normal Stress	0	-1.108

DIAGRAMS

Bending Moment Diagram



Shear Force Diagram



Displacement

Location (ft)	0	22.727	50
Total Deflection (in)	0	0.183	0
Deflection/Span	-	L/3284	-

The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the **model settings.**